

MOTHANA HASSAN AL-NUAIMI

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PROFILE SUMMARY

Artificial Intelligence graduate with hands-on experience in Machine Learning. Completing advanced AI engineering training through the AI.SPIRE Program, working on practical machine learning and AI projects. Proficient in building AI solutions using PyTorch, Hugging Face Transformers, and modern ML frameworks through academic research, internships, and independent projects. Practical exposure to AI-powered systems and data-driven applications through independently built projects, including a real-time sign language recognition system and a smart job-matching platform. Seeking an AI engineering or data science position to apply and further develop skills in deep learning, and intelligent system design.

SKILLS

- **Programming:** Python, SQL
- **AI & ML:** Machine Learning, NLP, Classification, RAG
- **Frameworks & Libraries:** PyTorch, Scikit-learn, OpenCV, Hugging Face Transformers
- **Data & Analytics:** SQL, Data Analysis, Data Preprocessing, Data Visualization
- **Tools:** Git, GitHub, Docker, Jupyter Notebook, VS Code, PyCharm
- **AI Techniques:** OCR, Information Extraction, Semantic Similarity, Image Processing, Model Evaluation • **Soft Skills:** Problem Solving • Communication • Research • Adaptability

PROJECTS

AI.SPIRE Applied Projects — NLP, Data & RAG

2026

- Completed applied AI engineering projects covering Python, SQL, Machine Learning, NLP, and RAG, with multiple test-based submissions including perfect scores of 18/18 and 9/9 on automated test suites.
- Built a sentiment classification pipeline and generated predictions for 1,033 app review samples, demonstrating end-to-end NLP delivery from raw text to inference output.
- Analyzed a student performance dataset of 2,000 records, identifying a 0.639 correlation between study hours and GPA; built an ETL pipeline processing 100 records that passed 3/3 validation tests.
- Built a PDF-based Retrieval-Augmented Generation (RAG) prototype: processed a 5-page document, split it into 11 searchable chunks, and indexed 384-dimensional embeddings using FAISS for semantic retrieval.

MayawmaJO — Full-Stack Job Platform

2026

A full-stack web platform designed to connect job seekers with employers for daily and part-time job opportunities in Jordan, supporting multiple user roles including job seekers, employers, and admins.

- Built core features including job search and application tracking for job seekers, job post creation and applicant management for employers, and a full admin dashboard for user management, employer verification, and platform monitoring.
- Integrated smart features such as AI-based job matching, fake job detection, trust scoring, and automated AI profile building to enhance platform reliability and user experience.
- **Technologies:** React, Node.js, Express.js, PostgreSQL, Prisma, JWT

DALILAK — Arabic Vehicle Manual RAG System

2026

- Developed an Arabic RAG pipeline processing 3 vehicle manuals (BYD, Geely, VW) using multilingual-e5-base embeddings; reduced average response time from 15s to under 5s.
- Built a 50+ question evaluation dataset across 6 vehicle system categories, achieving retrieval precision $\geq 90\%$ on a held-out test set.
- Deployed a GPU-accelerated Docker Compose stack with JWT authentication, bilingual Arabic/English UI, and page-level source citations.

Ishara — AI Sign Language Recognition System

2025

An AI-powered sign language recognition system that uses computer vision and deep learning to detect, process, and classify hand gestures from webcam video or image input in real time.

- Built a full pipeline including hand landmark extraction via MediaPipe (21 landmarks → 63 features), feature normalization and mirroring for left/right hand consistency, and classification using a PyTorch neural network with BatchNorm, ReLU, and Dropout layers.
- Developed a real-time webcam inference module and a Flask/Socket.IO server to stream live sign predictions to external interfaces, along with dataset preprocessing scripts using scikit-learn and Pandas.
- **Achieved 88% classification accuracy on a held-out test set.**
- **Technologies:** Python, PyTorch, OpenCV, MediaPipe, Flask, Socket.IO

EDUCATION

Bachelor's Degree in Artificial Intelligence — Al-Zaytoonah University of Jordan

Graduated: 2026

AI.SPIRE: Applied AI and ML Systems — LevelUp Economy & Istidama Consulting

2026

Python for AI - Applied NLP & Knowledge Systems

- 24-week intensive program covering applied NLP, machine learning, RAG systems, and Knowledge Graphs.
- Built and deployed end-to-end AI systems integrating NLP pipelines, vector search, and APIs.
- Developed professional skills in technical communication, teamwork, and AI system documentation.

COURSES & TRAINING

• **Introduction to Git and GitHub**

Google — Coursera | July 2026

• **Applied Data Science Track**

Qafza Training | 2026

• **Deep Neural Network for Beginners Using Python**

Packt — Coursera | May 2025

LANGUAGES

- **Arabic:** Native
- **English:** Strong technical English proficiency with a commitment to achieving full professional fluency.